Independence

Department of Statistical Sciences, University of Cape Town

Independence

- Definition: when the value of one observation does not influence or affect the value of another observation.
- They are genuinely separate and not influenced by each other.
- When are observations dependent?
    - Examples:
        - Repeated measurements on the same subject.
        - Measurements taken from subjects that are related.
        - Time series data where observations are collected over time and may be correlated.
- Typically occurs when experimental units are connected in some way
- and randomisation has not been implemented!

Independence in experimental design

- We can check for certain types of dependence with the residuals after model fitting.
- Beforehand, we can scrutinise the design of the study and look at the observations.

Dependence due to experimental design

- Applying treatments to the same group of students and measuring their performance.
- A study tests a new teaching method on several classrooms, and individual student test scores are used as the unit of analysis.
- Anything that indicates that experimental units are connected/related or could have been influenced by one another.

Spatial or temporal dependence

- When observations are collected over space or time, they may be correlated.
  - Measuring air pollution levels at different locations in a city. Locations closer to each other will have similar pollution levels.
  - Collecting daily temperature readings over a year. Temperatures on consecutive days are likely to be correlated.
- Dependence within a sample can occur when they are taken in a non-random sequence.
- Allows some other variable to introduce dependence between successive observations.

Example of temporal dependence

- Measure plant growth every morning, but you always measure Treatment A first, then Treatment B.
- If mornings get warmer as the sun rises (or you get tired and sloppy later in the day), then differences you see might be due to time of day, not the actual treatment.
- The danger: we might mistakenly think a treatment is effective, when actually it’s just correlated with the order of measurement.

How to check spatial or temporal dependence

- If we have the data on the spatial arrangement or time sequence of the observations, we can check for dependence using visualisations.
- If we assume that the order in which the data appear in the dataset = the order in which they were collected
- We can check for dependence using visualisations.

Visual checks for independence

- Plot the response/residuals against the order of data collection (or time).
- Look for patterns or trends…